

Future research with larger samples is needed, as sample sizes precluded investigation of several groups (less frequent racial categories, sexual and gender minorities, active-duty military) and intersections between factors. Future work should also examine contextual risk factors, including whether rates of violence differ across settings. Additionally, studies should examine perpetrator type with greater granularity: this study did not disaggregate provider/facilitator/guide, or separate partners from close others, and many participants reported the perpetrator falling outside these categories.

Overall, our findings suggest that a substantial proportion of people using psychedelics report experiencing sexual or physical violence while under the influence of the psychedelic, highlighting the need for further research, policy, harm reduction, and violence prevention efforts. Evidence-based prevention interventions (e.g., bystander training)⁷ should be tested and implemented in high-risk psychedelic use contexts (e.g., colleges, shelters). The study also identified key sociodemographic factors (unstable housing, students, young adults, people of colour) and mental health risk factors (substance use disorders, eating disorders, ADHD) that could inform tailored harm reduction and support resources.

Documenting safety outcomes of psychedelics – which are rapidly growing in popularity in the US, Europe, Australia, and beyond⁸ – in samples generalizable to the public is critical, and analogous studies should be conducted internationally. Risks with psyc-

delics are not exceptional, as sexual and physical violence victimization commonly involves substance use⁹. Whether psychedelics confer heightened or reduced risk relative to other substances, such as alcohol, cannabis or opioids, remains unknown. Nonetheless, the sizable proportion of participants reporting violence while under the effect of psychedelics underscores the need for safeguards and support across both supervised/therapeutic and recreational settings.

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The causality problem in climate anxiety research

Research investigating negative emotional responses to climate and environmental change – referred to variably as climate anxiety, eco-anxiety, climate worry, eco-distress, and solastalgia (hereafter “climate distress”) – is increasing rapidly¹. Although these responses have been recognized as understandable reactions to real and escalating global threats, they are also frequently portrayed as risk factors for mental health problems, including clinically significant psychological distress, anxiety, depression, and post-traumatic stress disorder (PTSD)²⁻⁴. Several recent systematic reviews suggest that eco-anxiety is related to psychological burden² or that solastalgia may mediate the onset or worsening of mental health problems⁴. However, the available evidence does not support these implied causal interpretations. To understand why, I examine the methodological and conceptual challenges that currently limit causal inference in climate distress research.

The first challenge is methodological. Nearly all studies included in existing reviews are cross-sectional quantitative surveys or qualitative/mixed-methods case studies²⁻⁴. These designs cannot establish temporality and therefore the causal direction. What they show is between-person correlations, but they do not capture whether increases in climate distress within a person lead to subsequent changes in mental health. To evaluate the within-person change necessary for causal inference, longitudinal or experimental studies are needed, but these are scarce.

Among existing longitudinal studies, associations with later

mental health are weak, often bidirectional, and highly sensitive to adjustment. For example, in New Zealand adults, higher climate concern was linked to slightly greater psychological distress one year later, but baseline distress also correlated with later concern, and the effects on life satisfaction were negligible once cross-lagged paths were modelled⁵.

Similar patterns appear in youth cohorts, where associations between climate worry and later depression, anxiety or sleep largely disappear after adjustment for prior mental health and psychosocial vulnerabilities⁶. Large population panels, such as the UK Household Longitudinal Study, likewise report either weak bidirectional associations or no evidence that climate worry causes later mental health problems⁷. Several well-powered, higher-quality studies report no significant long-term associations⁸.

The experimental evidence is even more limited. Existing studies typically examine short-term outcomes – such as state anxiety, hope or perceived agency – following brief informational manipulations, and none demonstrate sustained effects on generalized anxiety, depression, or other clinical outcomes.

Rather than climate distress causing mental health problems, the available evidence is more consistent with reverse causation. Individuals with pre-existing anxiety or depressive symptoms commonly exhibit heightened negative affectivity, attentional bias toward threats, and ruminative coping styles, making them more likely to respond with distress when confronted with a potential

stressor – whether a pandemic, nuclear war, or climate change.

This vulnerability pattern is well-documented in the broader research literature. Supporting this view, a recent large empirical study reports that “results point to prior distress possibly having a stronger impact on climate beliefs than beliefs have on distress”⁷. Furthermore, studies of dispositional factors, such as trait conscientiousness and antisocial behaviors, have been associated with stronger and weaker climate distress, respectively. These associations are more plausibly causal, since it is difficult to argue that low environmental concern itself produces antisocial behavior.

The second challenge that limits causal inference is conceptual. In much of the climate distress literature, the “exposure” is defined using indicators of emotional distress such as feelings of sadness, anxiety, or inability to cope – the same psychological concepts used to define mental health outcomes. When predictors and outcomes share this underlying cognitive or affective content, interpretation becomes circular, and distress appears to “cause” distress. Under these conditions, statistical mediation models are especially vulnerable to artefactual findings, because they cannot disentangle conceptually overlapping constructs.

This problem is evidenced empirically in a recent review on eco-anxiety and symptoms of major affective disorders, which notes that “stronger associations were observed where eco-anxiety was operationalized as ‘anxiety’ rather than ‘worry’”². More broadly, a synthesis of more than 200 studies shows that eco-anxiety is variably defined to encompass worry, fear, distress, anger, despair, and functional impairment, often without clear conceptual boundaries between these components¹. This conceptual ambiguity and overlap with mental health outcomes makes it unclear whether observed associations reflect distinct processes or artefacts of measurement.

The above issues are compounded by measurement practices. Most available instruments were developed for descriptive purposes rather than causal inference, and the most widely used scales⁹ – such as the Climate Change Anxiety Scale and the Hogg Eco-Anxiety Scale – combine cognitive worry (repetitive, future-oriented thought) with affective distress (fear, anxiety, sadness), behavioral responses, and, in some cases, functional impairment. Although this breadth helps reflect the complexity of climate-related emotions, it results in conceptual and empirical circularity.

Narrower measures, such as the Climate Change Worry Scale, avoid this problem by focusing on cognitive worry and affective intensity, which offers a cleaner definition of exposure. However, even these instruments lack adequate evidence of measurement invariance across age groups, cultural settings, and time, and few demonstrate sensitivity to within-person change, raising uncertainty about whether the observed differences reflect genuine psychological variation or scale artefacts.

Compounding these issues, many studies rely on single-item indicators (e.g., “How worried are you about climate change?”), use unvalidated measures, or apply them to populations for whom they have not been validated, such as children. Without clearer conceptual boundaries and measurement tools aligned with specific causal questions, observed associations between climate distress and mental health outcomes can only be interpreted as corre-

lates or markers, not as evidence of a distinct or causally operative psychological response to perceived environmental change.

A related challenge, alluded to above, is state-trait conflation. Many studies do not distinguish transient reactions from stable dispositions in either climate worry or psychological distress. A single item such as “I am deeply concerned about climate change” can capture both short-term fluctuations (e.g., following distressing climate news) and enduring characteristics such as low emotional stability, intolerance of uncertainty, or a general tendency to worry. Likewise, symptom scales with short recall windows (e.g., “past week”) may still reflect stable negative affectivity rather than situational mood. When measures of both climate distress and mental health contain substantial trait content, observed associations may primarily reflect shared vulnerability rather than a within-individual causal process in which climate distress drives later deterioration in mental health.

Taken together, current evidence does not support claims that climate distress causes mental health problems or worsens symptoms. Instead, the primary contribution of this work is documenting the extraordinary prevalence and heterogeneity of climate distress and related emotions, situating them as understandable and often morally grounded responses to profound and escalating global threats. It also helps identify a smaller subset of individuals for whom climate-related distress is persistent and functionally impairing, and who may benefit from targeted support, without treating their distress as a mental disorder or evidence of causal harm.

Given the public prominence of this topic, accurate framing matters. Even when authors caution that causality cannot be assumed, implied causal interpretations can move quickly from research literature into clinical and public discourse. In this sense, climate distress may be best understood as a signal of threat appraisal – sometimes adaptive, sometimes co-occurring with impairment – rather than evidence of harm in itself.

The central unresolved question is whether climate distress contributes to meaningful psychological harm. Addressing this issue requires research that aligns concepts, measurement, and causal design to within-person change; that clearly distinguishes cognitive concern, affective distress, and functional impairment; and that documents clinically meaningful outcomes. As climate and ecological challenges deepen, the call for clearer causal evidence will only grow.

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